Dallas International School

# Test 1 HL

**Name : . . . . . .**

**Topics 3.9 - 3.10 - 3.11 - 5.9**

**12th IB - Math AA**

**For HL Students**

**EXERCISE 1** Solve exactly for $0 \leq x \leq 2\pi$ :

**1.** $\sqrt{3}\sec 2x = -2$ [4]

**2.** $\cot x + 1 = 0$ [3]

**EXERCISE 2** Prove that $1 + \cot^2\theta = \text{cosec}^2\theta$. [4]

**EXERCISE 3** Show that : $(\cos\theta + \sin\theta)^2 + (\cos\theta - \sin\theta)^2 = 2$ [3]

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**EXERCISE 4** Consider the function $f(x) = 4\cot x + \sin x$, where $0 < x < \pi$.

**1. a.** Write $f(x)$ in terms of $\sin x$ and $\cos x$. [1]

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**b.** Hence, or otherwise, sketch the graph of $y = f(x)$, showing the value of the $x$-intercept. [2]

**2.** Find the value of $f^{-1}(2)$. [1]

It is given that $\sec\alpha = 1.5$, where $0 < \alpha < \pi$.

**3.** Find the value of $f(\alpha)$. [3]

## EXERCISE 5 BONUS

**1.** Show that $\cot 2\theta = \frac{1-\tan^2\theta}{2\tan\theta}$. [1]

**2.** Verify that $x = \tan\theta$ and $x = -\cot\theta$ satisfy the equation $x^2 + (2\cot 2\theta)x - 1 = 0$. [7]

**3.** Hence, or otherwise, show that the exact value of $\tan\frac{\pi}{12} = 2 - \sqrt{3}$. [5]

**4.** Using the results from parts (b) and (c), find the exact value of $\tan\frac{\pi}{24} - \cot\frac{\pi}{24}$.

Give your answer in the form $a + b\sqrt{3}$, where $a, b \in \mathbb{Z}$. [6]